

Markscheme

Mathematics: analysis and approaches
Practice question bank

103. (a)

It can be proven by contradiction that $\log_2 3$ is irrational (assume $\log_2 3 = \frac{p}{q}$ in lowest terms, then derive $2^p = 3^q$, which is impossible since one side is even and the other odd).

Given this, the equation $2^x = 3$ cannot have a rational solution. Find the value of x correct to 3 decimal places.

$$x = \log_2 3 = \frac{\ln 3}{\ln 2} \quad (M1)$$

$$= 1.585 \quad \text{A1}$$

[5 marks]